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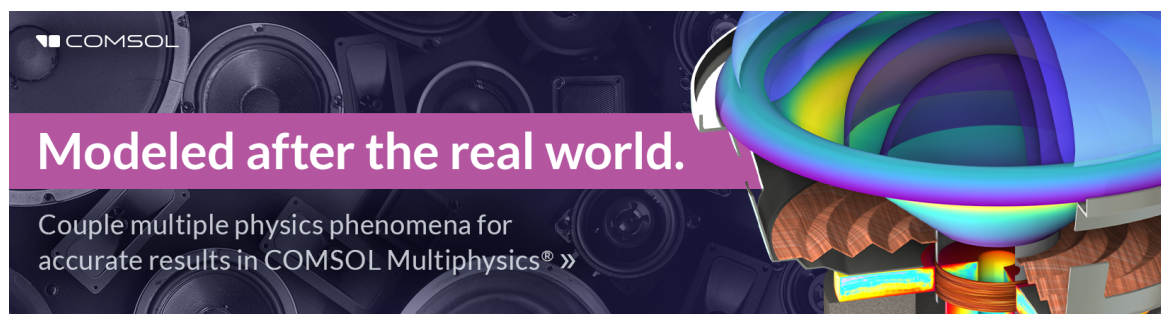


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An indirect method for the characterization of locally reacting liners

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An indirect technique for educing the homogenized acoustic impedance of a liner mounted on the wall of a barrel is presented. It is based on measurements and computational simulations of the multimodal scattering matrix of this lined duct. Measurements are performed with a multisource method and the use of an anechoic duct termination. The numerical computation of the scattering matrix relies on a finite element model, and assume that the duct is axisymmetric and uniformly covered by a locally reacting material. The impedance is educed by minimizing the difference between the theoretical and experimental acoustic power dissipations, which are deduced from the scattering matrix. The source is an incoming pressure field generated at one end of the duct only and composed of all cut-on modes. This technique was tested on a cylindrical barrel whose wall was partially lined with a realistic, locally reacting material made of honeycomb cells and a perforated facing sheet. Results for the acoustic impedance are compared with those deduced from semiempirical models and the experimental two microphone method. The best agreement with the indirect method is found with the semiempirical impedance results when the difference between the acoustic power dissipated by the actual lined barrel and the reference barrel is chosen as the cost function of the minimizing procedure.

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I. INTRODUCTION

The determination of the impedance of acoustic treatments placed in aircraft engines remains a challenge for the design of optimal liners and the reduction of the radiated noise. This is particularly true for experimental procedures since currently not any technique is able to fully reproduce, in a laboratory, the environment in the jet engine nacelle which the liner is submitted to.¹ The “two microphone” (TM) method^{2,3} is still commonly used because it represents a rapid and simple way to measure the impedance of a liner

sample by providing averaged value of the impedance over the area of the sample. But it is limited to normal plane wave incidence and no flow.

Another widely used method limited to locally reacting liners is the *in situ* technique introduced by Dean⁴ in 1974. It has been used extensively to measure the impedance with flow and it has recently been implemented on a full-scale engine test.¹ Usually liners are composed of a perforated sheet covering honeycomb cells, and are designed to be homogeneous, and thus modeled as a uniform impedance. But over a large area, this geometrical uniformity is generally not achieved due to the manufacturing processes and consequently the acoustic impedance cannot be considered uniform.^{5,6} Then a local technique does not seem to be the most appropriate¹ when one is concerned with realistic

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industrial liners.

In flow duct test facilities, the liner is mounted along the duct walls so as to allow for measurements that include the effect of the grazing flow along the liner. First, “classical” methods based on the measurement of the axial wave numbers^{7,8} in the lined section were developed assuming that a single mode propagates in the lined section. But this is not valid, for example, in a finite length liner where more complex acoustic fields are generated by the acoustic scattering at the liner edges. A method⁹ based on the determination of the eigenvalues of the lined cross section from the measurement of the pressure radial profile was also developed by solving the wall boundary condition to obtain the wall impedance.

Considerable effort has been put into providing analytical and numerical models for sound propagation in lined ducts with flow, and inverse techniques are now developed to determine the impedance of realistic liners. Liners for aeronautical applications are generally assumed to be locally reacting. To provide easier measurement conditions and to avoid problem implementing the liner in a cylindrical configuration, experimental facilities were constructed assuming incident plane wave in a hard rectangular cross section duct. Watson *et al.*^{10–12} developed a finite element based numerical method for educing the normalized impedance of a liner with an optimization algorithm that minimizes the difference between the measured and theoretical values of the acoustic pressure at the wall facing the liner. This method has been extended to flow with uniform^{13–15} and sheared^{16,17} profiles. Aurégan *et al.*¹⁸ proposed to couple an analytical model in the lined duct and measurements of the one-dimensional scattering matrix in the hard-wall ducts located on both sides of the lined section. A minimization criteria based on the matrix coefficients was performed to educe the impedance. But different liner impedances were found depending on the cost function defined from the matrix coefficients. Then Enaldy *et al.*^{19,20} proposed an analytical mode-matching model for the acoustic propagation in the lined section coupled with measurements to educe the liner impedance. Two cost functions were considered: (1) the sum of differences between measured and computed complex pressures at several duct positions, and (2) the difference between measured and calculated complex amplitudes of the scattered plane wave.

After comparing impedance data acquired during a multilaboratory study²¹ NASA Langley Research Center (NASA LaRC) commissioned a new set of experimental devices and a large-scale flow duct facility that is closer in size to a full-scale engine to extend the eduction method to three-dimensional modal distribution excitation.^{1,22} In parallel of these academic works, industrial companies such as Boeing,²³ General Electric,²⁴ and B.F. Goodrich²⁴ have developed indirect techniques to measure the acoustic impedance of aeronautic liners.

In this paper an inverse method performing the eduction of the homogenized impedance of a liner sample is proposed. It is based on a cost function deduced from the numerical and experimental multimodal acoustic power dissipations of the lined barrel. The dissipation is obtained from the numeri-

cal predictions and the measurements of the barrel multimodal scattering matrix. Due to the nature of the computational model involved, the problem presented in this paper assumes axisymmetry and no flow conditions with a locally reacting liner. But this method can be extended to more complex situations depending on the evolutions of the theoretical and experimental tools.

After presenting in Sec. II the definitions of the characteristic matrices of a duct and their physical interpretation, the theoretical basis for the measurements is introduced. Sections III and IV describe the experimental and numerical methods used to determine the scattering matrix of the lined barrel. In Sec. VI, the eduction technique is detailed after introducing in Sec. V how the liner dissipation can be deduced from the scattering matrix. Discussions of the experimental results for a locally reacting liner used in a cylindrical barrel are presented in Sec. VII.

II. THEORETICAL BASIS FOR THE EXPERIMENTAL TECHNIQUE

A. Expression for the acoustic pressure in the duct

Consider a cylindrical duct element between z_L and z_R with constant radius a (see Fig. 1). It is composed of a lined duct located between two semi-infinite rigid wall ducts. Assuming that the fluid is ideal, no mean flow and that linear acoustic theory applies, the acoustic pressure distribution in the hard duct sections is written in cylindrical coordinates (r, θ, z) for a frequency f as²⁵

$$p(r, \theta, z, t) = \sum_{m=-\infty}^{+\infty} \sum_{n=0}^{+\infty} P_{mn}(z) \Psi_{mn}(r, \theta) e^{-i\omega t}. \quad (1)$$

The subscripts m and n are, respectively, the azimuthal and radial mode numbers. P_{mn} are the modal pressure coefficients associated to the (m, n) mode and defined by

$$P_{mn}(z) = \frac{\langle p | \Psi_{mn} \rangle}{N_{mn}}, \quad (2)$$

where Ψ_{mn} are the eigenfunctions given by

$$\Psi_{mn}(r, \theta) = J_m(\chi_{mn} r/a) e^{im\theta}. \quad (3)$$

J_m is the Bessel function of the first kind of order m . χ_{mn} is the n th root satisfying the radial hard-wall boundary condition on the wall of the main duct [$J'(\chi_{mn}/a)=0$]. N_{mn} is the normalization constant associated to the (m, n) mode and defined by

$$N_{mn} = \langle \Psi_{mn} | \Psi_{mn} \rangle = S J_m^2(\chi_{mn}) \left[1 - \frac{m^2}{\chi_{mn}^2} \right], \quad (4)$$

$$N_{00} = 1,$$

where $S = \pi a^2$ is the cross section area. The total modal pressure coefficients $P_{mn}(z)$ in a cross section located at an axial coordinate z are given by the following relationship:

$$P_{mn}(z) = [P_{mn}^+(z) + P_{mn}^-(z)]. \quad (5)$$

Pressure waves traveling in the positive and negative z directions are denoted, respectively, in regions I and II by

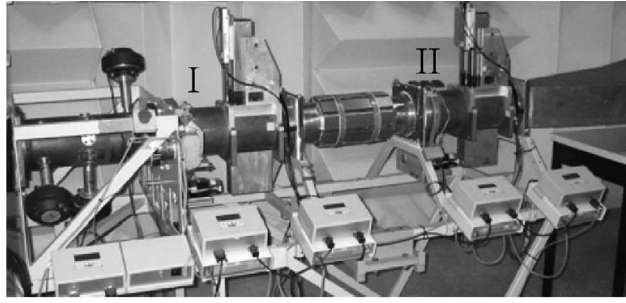


FIG. 1. View and schematic of the experimental setup.

$P_{mn}^{I\pm}(z) = P_{mn}^{I\pm}(0)e^{\pm ik_{mn}z}$ and $P_{mn}^{II\pm}(z) = P_{mn}^{II\pm}(0)e^{\pm ik_{mn}z}$. $k_{mn} = \sqrt{k^2 - (\chi_{mn}/a)^2}$ is the axial wave number of mode (m, n) in the main duct, $k = 2\pi f/c_0$, and c_0 is the speed of sound in air.

B. Definitions of the reflection, transmission, and scattering matrices of a duct element

The multimodal reflection matrix on side I, $[R^I]_{N \times N}$, of the duct element relates linearly the vector of the reflected modal pressures, $\{P^{I-}\}_N = \langle P_{00}^{I-}(z_L), \dots, P_{PQ}^{I-}(z_L) \rangle_N^T$ to the vector of the incident modal pressures $\{P^{I+}\}_N = \langle P_{00}^{I+}(z_L), \dots, P_{PQ}^{I+}(z_L) \rangle_N^T$ as follows:^{26,27}

$$\{P^{I-}\}_N = [R^I]_{N \times N} \{P^{I+}\}_N, \quad (6)$$

where N is the number of modes in both cross sections, P and Q are, respectively, the azimuthal and radial mode numbers associated to the N th propagating mode ($m \leq P$ and $n \leq Q$). This matrix characterizes the reflections on the left side of the duct element due to the liner and the termination.

The multimodal reflection matrix on side II of the same duct, $[R^{II}]_{N \times N}$ relates linearly the vector of the left-running modal pressures, $\{P^{II-}\}_N = \langle P_{00}^{II-}(z_R), \dots, P_{PQ}^{II-}(z_R) \rangle_N^T$ to the vector of the transmitted modal pressures $\{P^{II+}\}_N = \langle P_{00}^{II+}(z_R), \dots, P_{PQ}^{II+}(z_R) \rangle_N^T$ as follows:^{26,27}

$$\{P^{II-}\}_N = [R^{II}]_{N \times N} \{P^{II+}\}_N. \quad (7)$$

This matrix characterizes only the reflections from the termination. The multimodal transmission matrix $[T]_{N \times N}$ of the duct element relates linearly the vector of transmitted modal

pressures $\{P^{II+}\}_N = \langle P_{00}^{II+}(z_R), \dots, P_{PQ}^{II+}(z_R) \rangle_N^T$ to the vector of incident modal pressures $\{P^{I+}\}_N = \langle P_{00}^{I+}(z_L), \dots, P_{PQ}^{I+}(z_L) \rangle_N^T$ by the following expression:^{26,27}

$$\{P^{II+}\}_N = [T]_{N \times N} \{P^{I+}\}_N. \quad (8)$$

This matrix depends on the acoustic and geometric characteristics of the duct and the nature of the duct termination.

The scattering matrix $[S]_{N \times N}$ of the duct element relates the outgoing pressure wave vector $\{P^{out}\}_{2N} = \langle P_{00}^{I-}(z_L), \dots, P_{PQ}^{I-}(z_L), P_{00}^{II+}(z_R), \dots, P_{PQ}^{II+}(z_R) \rangle_{2N}^T$ to the incoming pressure wave vector $\{P^{in}\}_{2N} = \langle P_{00}^{I+}(z_L), \dots, P_{PQ}^{I+}(z_L), P_{00}^{II-}(z_R), \dots, P_{PQ}^{II-}(z_R) \rangle_{2N}^T$ (Fig. 1) as follows:^{28,29}

$$\begin{aligned} \{P^{out}\}_{2N} &= [S]_{2N \times 2N} \{P^{in}\}_{2N} \\ &= \begin{bmatrix} [S^{11}]_{N \times N} & [S^{12}]_{N \times N} \\ [S^{21}]_{N \times N} & [S^{22}]_{N \times N} \end{bmatrix}_{2N \times 2N} \{P^{in}\}_{2N}. \end{aligned} \quad (9)$$

This matrix represents an intrinsic characterization of the duct element independently of the upstream and downstream acoustic conditions: it depends only on acoustic and geometric features of the duct. It is composed by $(2N)^2$ coefficients $S_{mn,pq}^{kl}$; $k, l = 1, 2$ and $m, n, p, q = 1, 2, \dots, N$ and it provides a complete description of the modal reflection, transmission and mode conversion inside the element. The physical meaning of each elementary matrix is:

- $[S_{mn,pq}^{11}]_{N \times N}$ reflection of the wave entering the element from the left side,

- $[S_{mn,pq}^{21}]_{N \times N}$ transmission of the wave entering the element from the left side,
- $[S_{mn,pq}^{22}]_{N \times N}$ reflection of the wave entering the element from the right side, and
- $[S_{mn,pq}^{12}]_{N \times N}$ transmission of the wave entering the element from the right side.

For a given duct termination, these four matrices can be related as follows:

$$[R^I]_{N \times N} = [S^{11}]_{N \times N} + [S^{12}]_{N \times N} [R^{II}]_{N \times N} [T]_{N \times N}, \quad (10)$$

$$[T]_{N \times N} = [[I]_{N \times N} - [S^{22}]_{N \times N} [R^{II}]_{N \times N}]^{-1} [S^{21}]_{N \times N}, \quad (11)$$

where $[I]_{N \times N}$ is the identity matrix.

C. The procedure for measuring the scattering matrix

To measure the multimodal scattering matrix, a multi-load technique was developed by Sittel *et al.*³⁰ An anechoic termination is used,^{31,32} so as to reduce the duration of the experiment and assuming that the duct element is symmetric with respect to its median plan and to improve the results by reducing the stationary wave ratio. In fact, for this kind of termination, the acoustic reflections on side II are assumed to be zero ($[R_{\text{anechoic}}^{II}]_{N \times N} = [0]_{N \times N}$) and the relationships (10) and (11) become

$$[R_{\text{anechoic}}^I]_{N \times N} = [S^{11}]_{N \times N}, \quad (12)$$

$$[T_{\text{anechoic}}]_{N \times N} = [S^{21}]_{N \times N}. \quad (13)$$

By assuming symmetry with respect to the duct median plan, the duct reflection and transmission characteristics in both directions are identical:

$$[S^{11}]_{N \times N} = [S^{22}]_{N \times N}, \quad [S^{12}]_{N \times N} = [S^{21}]_{N \times N}. \quad (14)$$

Therefore when the termination is supposed to be anechoic the scattering matrix of duct element is deduced by measuring the reflection matrix on side I and the transmission matrix:

$$[S]_{2N \times 2N} = \begin{bmatrix} [R_{\text{anechoic}}^I]_{N \times N} & [T_{\text{anechoic}}]_{N \times N} \\ [T_{\text{anechoic}}]_{N \times N} & [R_{\text{anechoic}}^I]_{N \times N} \end{bmatrix}_{2N \times 2N}. \quad (15)$$

These matrices are measured with a technique³³ based on $M(M > N)$ source configurations to improve the experimental results. These configurations produce M linearly independent modal pressure distributions which are measured in both cross sections on either side of the duct element. After a modal decomposition procedure⁹ followed by a separation into incident and reflected modal pressure waves, the incoming and outgoing pressure vectors are obtained at z_L and z_R . These matrices are therefore deduced from

$$[T]_{N \times N} = [P^{II+}]_{N \times M} [P^{I+}]_{N \times M}^{-1}, \quad (16)$$

$$[R^I]_{N \times N} = [P^{I-}]_{N \times M} [P^{I+}]_{N \times M}^{-1}. \quad (17)$$

The lines of $[P^{I+}]$, $[P^{I-}]$, and $[P^{II+}]$ are composed, respectively, by the incident, reflected, and transmitted pressures corresponding to a given mode for the different source configurations. Their columns contain the acoustic modal pres-

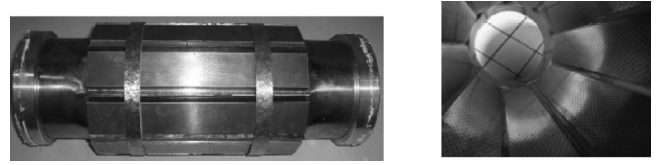


FIG. 2. Interior and exterior views of the barrel.

ures for a given measurement. The calculation of expressions (16) and (17) are performed by the MATLAB function *pinv*.

III. EXPERIMENTAL PROCEDURE

A. Hardware

The measurement of the scattering matrix $[S]$ was carried out with the experimental test rig (Fig. 1) built under a grant with E.E.C during the DUCAT program.³⁴ This facility and is installed in the anechoic chamber of the University of Technology of Compiègne, and is composed of all the necessary elements for the sound generation, measurement, and data treatment. The hardware equipment is made of five 0.5 m long cylindrical ducts with an internal radius $a=74$ and 10 mm thick steel walls except for the barrel wall which is made of 0.8 mm thick aluminum (Fig. 1):

- On the source section three acoustic drivers separated by 0.15 m along the z axis and by 120° in the azimuthal direction θ are flush mounted. The rotation of this section over 360° is controlled by motors and allows the variation of the source configuration. The source is connected to a white noise generator via a power amplifier.
- Each hard-wall duct elements (I) and (II) where measurements are made contains a B&K sound intensity probe pointing in the z direction. The distance between the $\frac{1}{4}$ in. microphones is 0.05 m according to the well-known conditions.^{9,27,35} To perform the measurements of the four pressure distributions $p(r, \theta)$ in the four cross sections, angular motors M_a rotate the ducts over 360° and radial motors M_r provide the radial displacement of the probe.
- The barrel whose middle segment consists of a 0.3 m long lined section is located between ducts (I) and (II). Views of the inside and outside of the barrel are shown in Fig. 2 and show the nine boxes which were built to assemble the Helmholtz resonators of the liner. Each aluminum box contains a honeycomb structure covered by a perforated facing sheet and five hard plates. Gaps between components are sealed to avoid acoustic leaks. Figure 2 shows the inside of the lined part of the barrel where nine strips are present. Previous works⁶ have shown that N_s strips cause the conversion of the incident azimuthal mode m into other azimuthal modes m' following the relationship $m' = m \pm \kappa \cdot N_s$, $\kappa=1, 2, \dots$. In the frequency range of our study and with $N_s=9$, the modal conversion will produce only evanescent modes and will therefore be neglected.
- The anechoic termination to avoid reflection on side II.

A work station operates automatically the rotation of the source and measurement sections, as well as the choice of the axial source and the displacements of the probe in the r

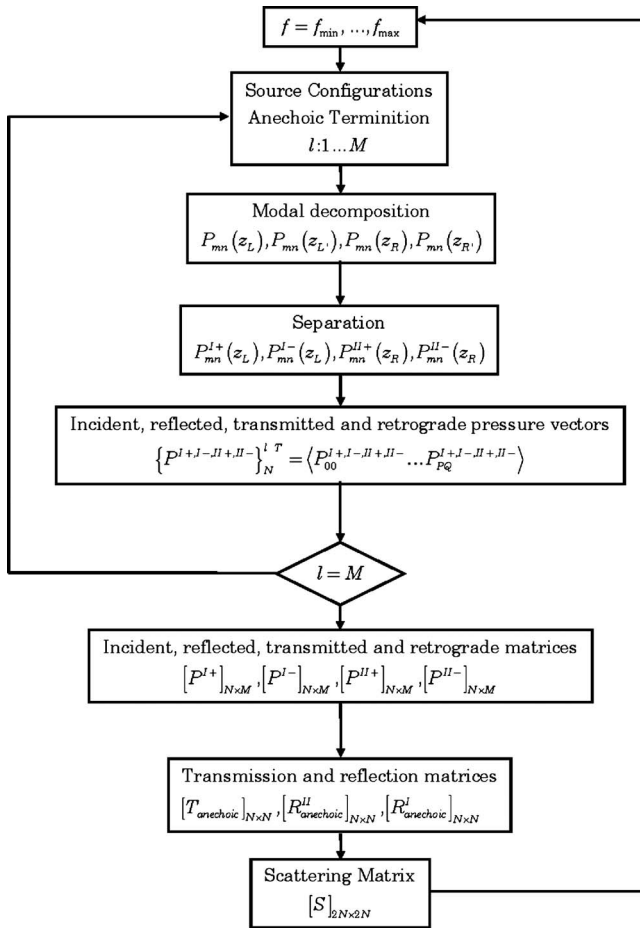


FIG. 3. Flow chart of the scattering matrix measurement.

direction. It also supplies the noise generation to the sources and perform the acquisition through an analog-digital converter for the signals from the four microphones.

B. Measurement procedure for the multimodal scattering matrix

The procedure to measure the scattering matrix is presented in Fig. 3. The acoustic source is driven with a white noise signal in the frequency range 0–2700 Hz leading to a Helmholtz number ka below 3.8 so that only $N \leq 5$ modes (0,0), $(\pm 1,0)$, and $(\pm 2,0)$ are cut-on. The transfer functions between the amplifier and microphone signals provide, after calibration, the amplitude and phase of the local acoustic pressure normalized by the level of the amplifier. The origin of the z axis is then given by the reference phase that is in the source z axis position. A 240 points mesh of the acoustic pressure field is measured in each of the two pairs of cross sections by rotating the measurement duct sections at 16 positions evenly spaced by 22.5° and by traversing the probe across 15 radial positions. Then a modal decomposition by a Fourier–Lommel’s transform⁹ for $m = -7$ to $+7$ and $n = 0-3$ is performed leading to the complex modal pressure coefficients in both pairs of cross sections. The incident and reflected modal vectors are then separated and computed³³ at z_L and z_R which are 1 m apart. This procedure is repeated for each source configuration. To compute the 100 coefficients of the matrix $[S]$, an overdetermination method with 12

TABLE I. Coordinates of the source configurations.

Configuration No.	θ_s^i (deg)	z_s^i (cm)
1	0	0
2	120	15
3	240	30
4	8	0
5	128	15
6	248	30
7	16	0
8	136	15
9	256	30
10	24	0
11	144	15
12	264	30

source configurations is chosen to ensure the generation of linearly independent input modal vectors.³⁰ The coordinates of the source configurations are given in Table I.

Finally the matrices $[P^{\text{in}}]_{2N \times 2N}$ and $[P^{\text{out}}]_{2N \times 2N}$ are filled with the modal coefficients vectors $\{P_{mn}^{I+}(z_L)\}$, $\{P_{mn}^{II+}(z_R)\}$, $\{P_{mn}^{I-}(z_L)\}$, and $\{P_{mn}^{II-}(z_R)\}$. The scattering matrix is then obtained by taking into account only the propagating modes, i.e., by assuming that the influence of the evanescent modes can be neglected.

IV. NUMERICAL COMPUTATION OF THE MULTIMODAL SCATTERING MATRIX

A. Theoretical formulation

For the purpose of computing numerically its scattering matrix, the duct element located between z_L and z_R is assumed axisymmetric, as shown in Fig. 4. As the experimental duct, it consists of a lined duct between two hard-wall ducts. The surface Γ_{WD} with its normal vector $\vec{n}_{\text{WD}}$ represents the rigid walls, Γ_{LD} with its normal vector $\vec{n}_{\text{LD}}$ represents the lined surface. Ω denotes the acoustic domain inside the duct. The liner is assumed to be locally reacting and is defined by its normalized acoustic impedance Z which depends only on frequency. The acoustic pressure inside the duct is solution of the Helmholtz equation together with the boundary conditions at Γ_{WD} and Γ_{LD} :

$$k^2 p + \Delta p = 0 \quad (\Omega),$$

$$\frac{\partial p}{\partial n_{\text{WD}}} = 0 \quad (\Gamma_{\text{WD}}),$$

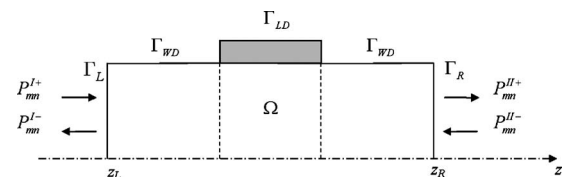


FIG. 4. Schematic of the theoretical model.

$$Z \frac{\partial p}{\partial n_{LD}} = i\omega \rho p \quad (\Gamma_{LD}). \quad (18)$$

Because of the axisymmetry, the acoustic field is written for a given azimuthal order m :

$$p(r, \theta, z) = \sum_{m=-\infty}^{+\infty} p_m(r, z) e^{im\theta}. \quad (19)$$

The incoming and outgoing pressure waves at Γ_L and Γ_R located, respectively, at z_L and z_R with N_r the number of radial modes are then given by

$$p_m^{(L)} = \sum_n^{N_r} (P_{mn}^{I+} e^{ik_{mn}(z-z_L)} + P_{mn}^{I-} e^{-ik_{mn}(z-z_L)}) J_m(\chi_{mn} r), \quad (20)$$

$$p_m^{(R)} = \sum_n^{N_r} (P_{mn}^{II+} e^{ik_{mn}(z-z_R)} + P_{mn}^{II-} e^{-ik_{mn}(z-z_R)}) J_m(\chi_{mn} r). \quad (21)$$

B. The finite element method

To solve this boundary-value problem, the standard finite element method is used. The Helmholtz equation is written as a weak variational formulation:

$$\Pi = - \int_{\Omega} (\nabla q \cdot \nabla p) d\Omega + k^2 \int_{\Omega} qp d\Omega + \int_{\cup \Gamma_i} q \frac{\partial p}{\partial n} d\Gamma = 0, \quad (22)$$

where q is the test function. The last integral in this formulation is given by the following expression:

$$\begin{aligned} \int_{\cup \Gamma_i} q \frac{\partial p}{\partial n} d\Gamma = & \int_{\Gamma_L} q \left(\frac{i\omega \rho p}{Z} \right) d\Gamma_{LD} + \sum_n^{N_r} ik_{mn} \left[\left(n_L (P_{mn}^{I+} \right. \right. \\ & - P_{mn}^{I-}) \int_{\Gamma_L} q J_m(\chi_{mn} r) d\Gamma_L \Big) + \left(n_R (P_{mn}^{II+} \right. \\ & - P_{mn}^{II-}) \int_{\Gamma_R} q J_m(\chi_{mn} r) d\Gamma_R \Big) \Big]. \end{aligned} \quad (23)$$

$\vec{n}_L$ and $\vec{n}_R$ are the normal vectors, respectively, to Γ_L and Γ_R . The use of the modal decompositions at Γ_L and Γ_R introduces the modal amplitudes as additional degrees of freedom to the model. It is therefore necessary to complete Eq. (22) with additional equations to have a well-posed problem. This is done with the hypothesis that the modal amplitudes at Γ_L and Γ_R can be obtained by the projection of the acoustic field on the eigenfunctions of the hard-wall duct:

$$\int_{\Gamma_L} p J_m(\chi_{mn} r) d\Gamma_L = (P_{mn}^{I+} + P_{mn}^{I-}) \int_{\Gamma_L} J_m(\chi_{mn} r)^2 d\Gamma_L,$$

$$\int_{\Gamma_R} p J_m(\chi_{mn} r) d\Gamma_R = (P_{mn}^{II+} + P_{mn}^{II-}) \int_{\Gamma_R} J_m(\chi_{mn} r)^2 d\Gamma_R. \quad (24)$$

To solve Eqs. (22) and (24), the domain Ω is meshed with triangular linear elements,³⁶ and the solution is also approximated by a linear interpolation.

C. Computation of the multimodal scattering matrix

For a given azimuthal order m , and after taking into account the incoming and outgoing modal pressures on each boundary Γ_L and Γ_R , the arrangement of the system (22) results in the following algebraic system:

$$\begin{aligned} & \langle \langle q_1 \cdots q_M \rangle_M \langle 1 \cdots 1 \rangle \rangle \\ & \times \begin{bmatrix} [K]_{M \times M} & [E_1]_{M \times N_r} & [E_2]_{M \times N_r} & [F_1]_{M \times N_r} & [F_2]_{M \times N_r} \\ [G_1]_{N_r \times M} & [G_2]_{N_r \times N_r} & [G_3]_{N_r \times N_r} & [0] & [0] \\ [0] & [0] & [0] & [0] & [0] \\ [0] & [0] & [0] & [0] & [0] \\ [H_1]_{N_r \times M} & [0] & [0] & [H_2]_{N_r \times N} & [H_3]_{N_r \times N} \end{bmatrix} \\ & \times \begin{bmatrix} \begin{Bmatrix} p_1 \\ \vdots \\ p_M \end{Bmatrix}_M \\ \{P_{mn}^{I-}\}_{N_r} \\ \{P_{mn}^{I+}\}_{N_r} \\ \{P_{mn}^{II-}\}_{N_r} \\ \{P_{mn}^{II+}\}_{N_r} \end{bmatrix} = \{0\}, \end{aligned} \quad (25)$$

where M is the node number in the domain Ω , $[K]$ relates the test function to the nodal pressures in the domain, and $[E_1]$, $[E_2]$, $[F_1]$, and $[F_2]$ relate the test functions to modal pressures on Γ_L and Γ_R . $[G_1]$, $[G_2]$, and $[G_3]$ relate the nodal acoustic pressures in Ω to modal pressures on the boundary Γ_L (P_{mn}^{I-} , P_{mn}^{I+}). Finally $[H_1]$, $[H_2]$, and $[H_3]$ relate the nodal acoustic pressures to different modal pressures on the boundary Γ_R (P_{mn}^{II-} , P_{mn}^{II+}). These matrices are directly obtained from Eqs. (22) and (24).

For a given m , the azimuthal scattering matrix is defined as

$$\begin{Bmatrix} \{P_{mn}^{I-}\} \\ \{P_{mn}^{II+}\} \end{Bmatrix} = [\Delta]_{2N_r \times 2N_r} \begin{Bmatrix} \{P_{mn}^{I+}\} \\ \{P_{mn}^{II-}\} \end{Bmatrix}. \quad (26)$$

This matrix can be obtained by written the system of Eq. (25) as follows:

$$\begin{aligned} & [K]\{p\} + [A] \begin{Bmatrix} \{P_{mn}^{I+}\} \\ \{P_{mn}^{II-}\} \end{Bmatrix} + [B] \begin{Bmatrix} \{P_{mn}^{I-}\} \\ \{P_{mn}^{II+}\} \end{Bmatrix} = \{0\}, \\ & [C]\{p\} + [U] \begin{Bmatrix} \{P_{mn}^{I+}\} \\ \{P_{mn}^{II-}\} \end{Bmatrix} + [V] \begin{Bmatrix} \{P_{mn}^{I-}\} \\ \{P_{mn}^{II+}\} \end{Bmatrix} = \{0\}, \end{aligned} \quad (27)$$

where $\{p\}$ is the nodal acoustic pressure vector and the matrices $[A]$, $[B]$, $[C]$, $[U]$, and $[V]$ are defined as

$$[A] = [[E_1][F_2]], \quad [B] = [[E_2][F_1]], \quad [C] = [G_1] + [H_1],$$

$$[U] = [[G_2][H_3]], \quad [V] = [[G_3][H_2]]. \quad (28)$$

The azimuthal scattering matrix is written as

$$[\Delta] = ([V] - [C][K]^{-1}[B]^{-1})([U] - [C][K]^{-1}[A]^{-1}). \quad (29)$$

The total scattering matrix $[S]_{2N \times 2N}$ is obtained by repeating this operation for each m and by gathering the azimuthal matrices $[\Delta]_{2N_f \times 2N_f}$.

V. COMPUTATION OF THE ACOUSTIC POWER DISSIPATION

The acoustic power dissipation W_{dis} of a duct is defined as the difference between the total acoustic power of incoming waves W^{in} and the total acoustic power of outgoing waves W^{out} :

$$W_{\text{dis}} = W^{\text{in}} - W^{\text{out}}. \quad (30)$$

The acoustic power dissipation can be written as (see Appendix)

$$W_{\text{dis}}(\text{Watt}) = \sum_{i=1}^{2N} (1 - \lambda_i) |d_i|^2$$

$$= \sum_{i=1}^{2N} \left[(1 - \lambda_i) \left| \sum_{j=1}^{2N} (TR_{ij} P_j^{\text{in}}) \right|^2 \right] \quad (31)$$

and

$$W_{\text{dis}}(\text{dB}) = 10 \log_{10} \left(\frac{W_{\text{dis}}(\text{Watt})}{10^{-12}} \right). \quad (32)$$

Equation (31) indicates that the overall acoustic power dissipation depends on the acoustic and geometrical properties of the duct element defined by $[TR]_{2N \times 2N} = [U]^* T [X]$ and on the incident vector $\{P^{\text{in}}\}$.

VI. THE INDIRECT METHOD TO EDUCE THE ACOUSTIC IMPEDANCE

The liner normalized acoustic impedance is educed for each frequency by minimizing the cost function defined as the difference between the measured and computed acoustic power dissipations:

$$F_{\text{obj}} = |W_{\text{exp dis}}(\text{Watt}) - W_{\text{num dis}}(\text{Watt})|. \quad (33)$$

The minimization is performed by the MATLAB function “fminsearch” based on the Nelder–Mead’s simplex method³⁷ by iteratively adjusting the impedance value. It is initialized by the normalized acoustic impedance given by a semiempirical model^{38,39} so as to improve the convergence of the process. As explained in Sec. V, the acoustic power dissipation is deduced from both the experimental and numerical scattering matrices of the 1 m long duct element located between z_L and z_R , but it also depends on the incident modal pressure vector.

In a previous work,³⁰ experimental results of the scattering matrix coefficients of a 0.01 m thick steel duct have shown that around the mode cut-on frequencies the hard-wall duct condition was not verified, leading to acoustic power

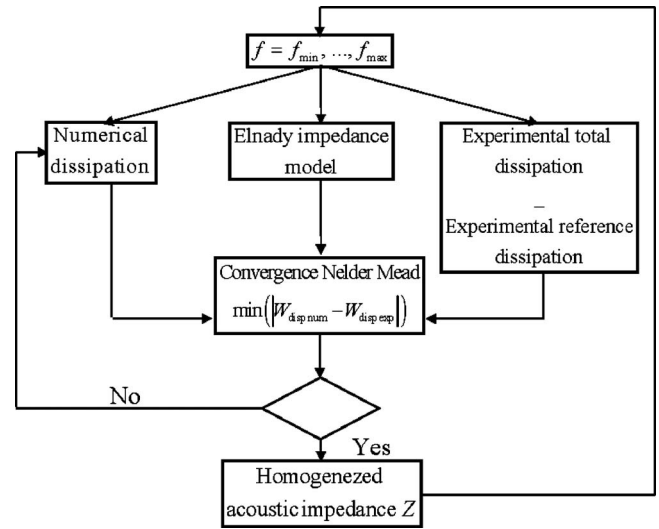


FIG. 5. Flow chart of the indirect method to educe the homogenized acoustic impedance.

dissipation. Therefore the measured acoustic power dissipation for a given liner is assumed to be the sum of the dissipation effectively produced by the liner and the dissipation caused by a reference duct that does not fully satisfy the hard-wall condition:

$$W_{\text{exp dis}}(\text{Watt}) = W_{\text{ref exp dis}}(\text{Watt}) + W_{\text{liner exp dis}}(\text{Watt}). \quad (34)$$

The flow chart of the indirect method is presented in Fig. 5 and indicates that the impedance which will come out of procedure will be that of an equivalent material covering the lined surface of the barrel averaged over an incident pressure wave vector distributed over all the cut-on modes.

VII. ANALYSIS OF RESULTS

A. The duct element under test

As already described the duct element to test is located between the axial positions z_R and z_L (Fig. 1) and is symmetric with respect to the duct axis and to the middle of the duct element. The experiment is conducted with a barrel (Fig. 2) equipped with a locally reacting liner composed of a perforated facing sheet (thickness $e=0.8$ mm, hole diameter $D=1$ mm, and open area $\sigma=5\%$), honeycomb cells (depth $L=20$ mm), and backed by a solid plate. The resonance frequency of the Helmholtz resonators is $ka=2.22$ according to the semiempirical model.³⁸ As explained in Sec. VI, to measure $W_{\text{ref exp dis}}$ an additional experiment called “reference” has to be carried out with the same barrel but with the perforated plate covered with a metallic sheet and the honeycomb removed to remove the dissipation effect of the liner and leaving only the barrel skin.

B. The scattering matrix coefficients

In this section, the experimental results for the scattering matrix coefficients are compared against numerical data obtained with the normalized acoustic impedance resulting from two semiempirical models^{38,40,41} and an experiment

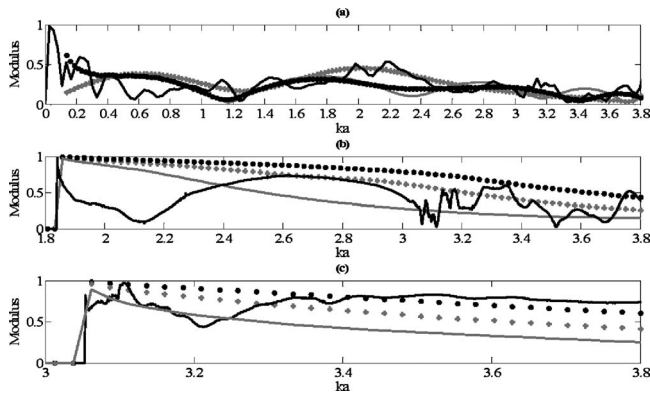


FIG. 6. The experimental modulus (black line) of scattering matrix coefficients [(a) $S_{00,00}^{11}$, (b) $S_{-10,-10}^{11}$, and (c) $S_{-20,-20}^{11}$] of the duct element compared with the numerical result deduced with theoretical (Elnady: gray line; Guess: *) and TM method impedances (●).

based upon the TM method.³ In Fig. 6(a), the experimental and theoretical scattering coefficients $|S_{00,00}^{21}|$ are plotted against ka . The experimental transmission coefficient of the plane wave is shown to be close to 1 up to $ka=1.2$ where it decreases to a point where there is no transmission in the frequency band $ka=[2-2.5]$ where the liner theoretical resonance is found. Then, $|S_{00,00}^{21}|$ increases but remains lower than 0.8. In the low frequency band up to $ka=0.4$ the separation step causes discrepancy on the result as expected.³¹ The experimental results agree with theory up to $ka=2.5$ only when the Elnady's impedance modeling is used. Indeed, even the minima of transmission are not correctly predicted by the numerical method with the impedance values from the experiments and from Guess's model. In Figs. 6(b) and 6(c) the experimental and theoretical scattering coefficients $|S_{-10,-10}^{21}|$ and $|S_{-20,-20}^{21}|$ of, respectively, modes $(-1,0)$ and $(-2,0)$ are plotted versus ka . For both higher-order modes, the experimental transmission remains close to zero except for the mode $(-1,0)$ and for frequencies higher than $ka=3.1$. A comparison with theory shows again good agreement with experiments when Elnady's model of impedance is used but only for the mode $(-1,0)$. In fact, other numerical computations lead to coefficients close to zero. In Fig. 7(a)–7(c), the modulus of the experimental and theoretical scattering coefficients $|S_{00,00}^{11}|$, $|S_{-10,-10}^{11}|$, and $|S_{-20,-20}^{11}|$ are

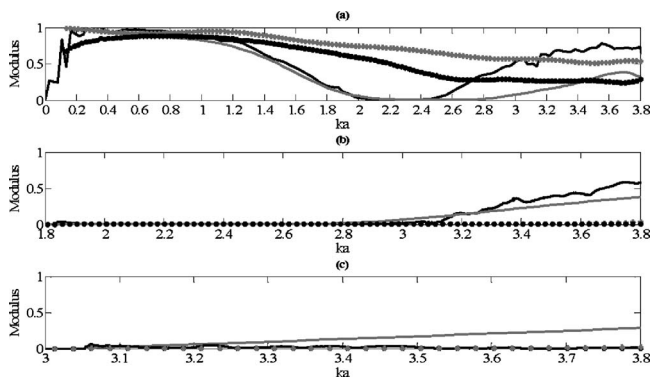


FIG. 7. The experimental modulus (black line) of scattering matrix coefficients [(a) $S_{00,00}^{21}$, (b) $S_{-10,-10}^{21}$, and (c) $S_{-20,-20}^{21}$] of the duct element compared with the numerical result deduced with theoretical (Elnady: gray line; Guess: *) and TM method impedances (●).

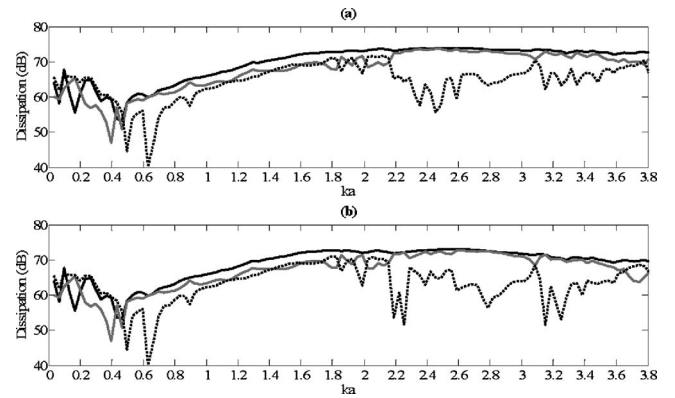


FIG. 8. Experimental acoustic power dissipations of the duct element and reference duct (a) in presence of all modes and (b) for the plane wave excitation: (black line) total, (gray line) liner, and (dashed line) reference.

plotted versus ka , they represent, respectively, the reflection of the plane wave, and of the $(-1,0)$ and $(-2,0)$ modes. Unlike the transmission coefficients, oscillations on the experimental data which were previously³⁰ shown to be due to the high sensitivity of the reflection coefficients to uncertainties in temperature and modal pressure measurements are apparent in Fig. 7(a) but also in Figs. 7(b) and 7(c) near the cut-on frequencies. In Fig. 7(b) the graph of $|S_{-10,-10}^{11}|$ indicates an important disagreement with theory especially in the frequency band around $ka=2.2$ where a minimum is measured at 0.1 instead of 0.8, 0.9, and 0.92 as numerically predicted with impedances, respectively, calculated from the Elnady's or Guess's models and measured by the TM method.

The results presented so far show great differences between numerical scattering matrix coefficients calculated with impedance values based on two models and an experiment. This illustrates how difficult it is to predict the impedance of a typical perforate liner. Elnady's model seems to be the more appropriate to represent the liner mounted on the barrel and will therefore be used to initialize the iteration during the minimization process. But the agreement with this model is valid only when a plane wave is incident. Indeed, the discrepancies observed when higher-order modes are incident show that specific effects are produced, thus justifying conducting an additional experiment with a reference duct.

C. Acoustic power dissipation

The acoustic power dissipation of the barrel depends on the liner properties and on the incident pressure wave vector [Eq. (32)]. The acoustic power dissipations measured for the reference and lined cases are therefore compared in Figs. 8(a), 8(b), 9(a), and 9(b) versus ka for four different incident pressure wave vectors entering the duct element from the left side $\{P_{2N}^{\text{in}}\}^T$:

- All modes with unit amplitude and in phase: $\{P_{2N}^{\text{in}}\} = \langle 1 \ 1 \ 1 \ 1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0 \rangle^T$ in Fig. 8(a).
- Plane wave: $\{P_{2N}^{\text{in}}\} = \langle 1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \rangle^T$ in Fig. 8(b).
- Mode $(1,0)$: $\{P_{2N}^{\text{in}}\} = \langle 0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \rangle^T$ in Fig. 9(a).

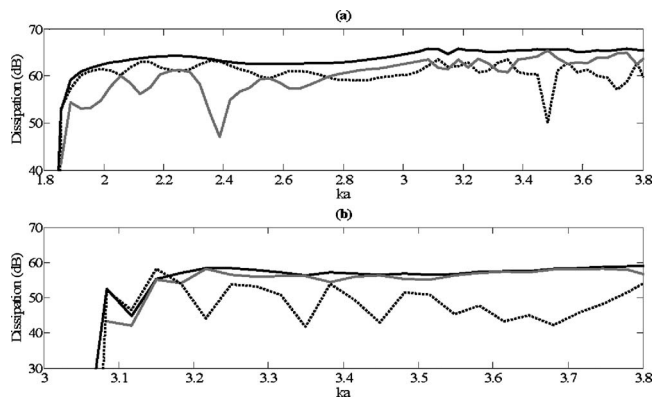


FIG. 9. Experimental acoustic power dissipations of the duct element and reference duct (a) for the mode (1,0) excitation and (b) for the mode (2,0) excitation: (black line) total, (gray line) liner, and (dashed line) reference.

- Mode (2,0): $\{P^{\text{in}}\}_{2N} = \langle 0 \ 0 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \rangle^T$ in Fig. 9(b).

For the first and second cases dissipation levels are, respectively, 8 and 13 dB higher than the dissipations of modes (1,0) and (2,0). This result is expected because in the second case makes the acoustic power carried by the higher modes is lower than that of the plane wave.³¹ For the first and second cases, the dissipation increases with ka reaching a maximum of 72 dB around the resonance frequency of the liner. In Fig. 8(a) the comparison of the three acoustic power dissipations measured in the presence of all modes shows that in the frequency bands $ka=[0.4-1.4]$ and $ka=[2.2-3.8]$ the liner is dissipating more than the reference by 2–20 dB. In the low frequency range, where uncertainties are important as explained in Sec. VI, and the liner is assumed to be inefficient the total dissipation is resulting of the “hard-wall condition.” In the frequency band $ka=[1.4-2.2]$ which contains the cut-on frequency of the mode (1,0), the liner and reference dissipations are similar. A strong coupling effect between acoustic and structural modes are expected as already observed in a previous work.³¹

This remark is emphasized by considering $|S_{-10,-10}^{11}|$ the (1,0) reflection coefficient [Fig. 7(b)] and at the reference acoustic power dissipation which is up to 15 dB higher in this frequency band than the liner dissipation [Fig. 9(a)]. The results for acoustic power dissipations produced with the incident mode (2,0) are plotted in Fig. 9(b). They show a 60 dB power dissipation much lower than the plane wave. Also in this high frequency band the reference dissipation remains about 10 dB lower than the total except near the cut-on frequency where the uncertainties are important.³¹

This analysis shows that the effect of the assembling conditions on the prediction of the liner impedance has to be taken into account especially in the frequency band $ka=[1.8-2.2]$. Therefore to educe the homogenized acoustic normalized impedance the indirect technique will be based on the acoustic power dissipation of the liner “alone” as formulated in Eq. (34).

D. Acoustic impedances

In Figs. 10(a) and 10(b) the real and imaginary parts of the educed liner normalized impedance computed in pres-

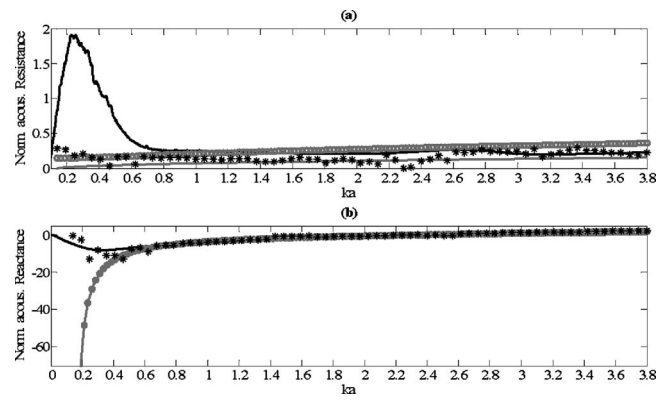


FIG. 10. (a) The educed normalized acoustic resistance, (b) The educed normalized acoustic reactance of the liner (*) compared with results from semitheories (Elnady: $\circ$; Guess: gray line) and TM method (black line).

ence of all the cut-on modes are compared with the theoretical^{38,40} and experimental impedances, respectively. The latter is measured with the TM^{2,3} method assuming plane wave propagation on two 0.01 and 0.003 m diameter samples manufactured with the same specifications as the liner located on the barrel. The comparison shows that TM method and both theoretical models lead to impedance values varying within a band of about 0.1 and 1 for, respectively, the resistance and reactance. The TM method values are mostly located between Guess’s model⁴⁰ lower limit and Elnady’s model³⁸ upper limit, except in the low frequency range where TM method real and imaginary parts are much higher, as observed in the literature.¹

Except in the frequency domain around $ka=2.3$ where the real part is close to zero, the educed resistances are displaying smooth curves located between the two theoretical curves. The educed imaginary part is found to be close to Elnady’s model³⁸ prediction except near $ka=2$. The zeros of the imaginary parts, corresponding to the liner resonance frequency, are also varying. The educed value is found equal to $ka=2.45$ close to Elnady’s model³⁸ prediction $ka=2.22$ but far from Guess’s model⁴⁰ value $ka=3.30$ and from the experimental TM method value $ka=2.95$.

Results of the educed impedance per mode presented in Figs. 11(a) and 11(b) show that when the incident pressure

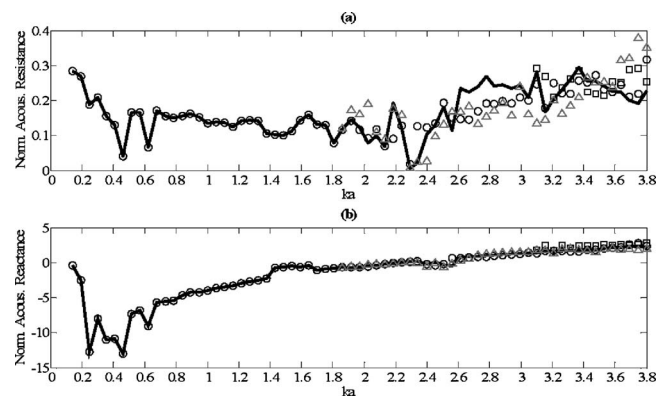


FIG. 11. (a) Comparison of the educed normalized acoustic resistances. (b) Comparison of the educed normalized acoustic reactances of the liner for single mode and all cut-on modes incident pressure vectors: [all modes: (—); (0,0): $\circ$; (1,0): $\triangle$; (2,0): $\square$].

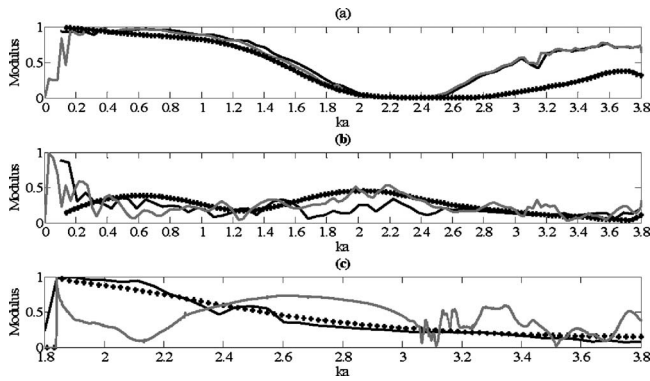


FIG. 12. Comparison of the scattering matrix coefficients: (a) $S_{00,00}^{21}$, (b) $S_{00,00}^{11}$, and (c) $S_{-10,-10}^{11}$ numerically recalculated with the educed impedance (black line) and with Elnady's semitheoretical impedance (*) and experimental results (gray line).

vector is modified, the educed impedance values are found to vary but within the limits of 0.1 and 1, respectively, for the real and imaginary parts. These uncertainties are probably due to the measurement errors, to the strong hypothesis that the dissipations produced by the liner and the reference configuration are uncoupled and to the nonuniformity of the wall surface due to the mounting conditions with the presence of the strips. The educed impedance can therefore be considered not to depend on the modal incident pressure vector verifying the locally reacting property of the liner. The educed impedance method applied with all the cut-on modes as incident pressure vector allows therefore to reduce these uncertainties leading then to a homogenized impedance.

E. Recalculating the scattering matrix

To verify the validity of the hypothesis formulated in Eq. (34) about the dissipations, the scattering matrix coefficients $|S_{00,00}^{21}|$, $|S_{00,00}^{11}|$, and $|S_{-10,-10}^{11}|$ are numerically computed again but with the educed acoustic impedance obtained with all the cut-on modes for the incident pressure wave. The results are compared in Figs. 12(a)–12(c), with numerical results performed with Elnady's model and TM method impedances and with the experimental results achieved with the lined barrel. The modulus of the plane wave transmission coefficient $|S_{00,00}^{21}|$ computed with the theoretical impedances plotted in Fig. 12(a) is very similar to the experimental curves and to the curves obtained with Elnady's model but up to $ka=2.4$. This figure also shows that the transmission of the plane wave mode is not sensitive to the departure from the hard-wall condition with a nonzero acoustic particle velocity. This remark can also be made regarding the results for the reflection coefficient in Fig. 12(b) despite the uncertainties which affect the results.³⁰ In Fig. 12(c) the plot of the mode $(-1,0)$ reflection coefficient with the educed impedance is shown to be similar to the plot deduced with Elnady's impedance. This result indicates that the reflection of this higher-order mode is very sensitive to the wall condition of the barrel and that the approximate model which assumes the superposition of the dissipations [Eq. (34)] coming from the liner and from the reference is acceptable.

VIII. CONCLUSIONS

In this work an indirect method based on the minimization of the experimental and numerical acoustic power dissipations for a multimodal incident pressure wave vector is performed to educe the homogenized acoustic impedance of a liner located on a cylindrical barrel. The numerical model assumes axisymmetric and locally reacting liners. The experimental acoustic power dissipated by the material is assumed to be the difference between the dissipations of the lined barrel and of a reference barrel, both being deduced from the measurement of their respective scattering matrices.

This indirect technique was tested with a standard perforate liner. The educed acoustic impedance was compared with results given by models and measured with the TM method which among them already lead to large differences in the impedances and in the liner resonance frequencies.

The impedance eduction was conducted for three incident pressure wave vectors targeting one mode at a time. Taking into account the uncertainties, no obvious dependence of the educed impedance on the modal vector could be found, thus verifying the locally reacting property of the lined wall. The educed impedance which is deduced for an incident pressure vector with equal modal amplitudes from the numerical dissipation of a uniform and locally reacting wall duct and the experimental dissipation where the reference duct contribution was removed, results in a homogenized impedance fitting the imperfections of the material with its mounting condition and reducing the effects of the nonrespect of the zero particle velocity on parts of the barrel. This hypothesis is validated by the good agreement found during the comparison between scattering coefficients numerically computed with Elnady's impedance model and the educed impedance deduced with the all mode incident pressure wave vector. The next step of the study will be to improve this technique in order to reduce the sources of error and to extend it to cases with mean grazing flow.

ACKNOWLEDGEMENTS

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NOMENCLATURE

z_L, z_R	The left and right axial coordinates of the duct under test
a	Duct radius
(r, θ, z)	Cylindrical coordinates
f	Frequency
t	Time
p	Acoustic pressure
m	Azimuthal mode number
n	Radial mode number
P_{mn}	Modal pressure coefficient associated to the (m, n) mode
Ψ_{mn}	The eigenfunctions
J_m	The Bessel function of the first kind of order m
χ_{mn}	The n th root of the derivative of J_m

N_{mn} = The normalization constant associated to the (m, n) mode
 $S = \pi a^2$ = The cross section area
 $k_{mn} = \sqrt{k^2 - (\chi_{mn}/a)^2}$ = The axial wave number of mode (m, n)
 $k = 2\pi f/c_0$ = Total wave number
 ka = Total adimensional wave number
 c_0 = The speed of sound in air
 $[R^I]_{N \times N}$ = The multimodal reflection matrix on side I
 $\{P^I\}_N$ = The vector of the reflected modal pressures
 $\{P^{I+}\}_N$ = The vector of the incident modal pressures
 N = The number of modes
 P = The azimuthal mode number associated to the N th propagating mode
 Q = The radial mode number associated to the N th propagating mode
 $[R^{II}]_{N \times N}$ = The multimodal reflection matrix on side II
 $\{P^{II}\}_N$ = The vector of the left-running modal pressures
 $\{P^{II+}\}_N$ = The vector of the transmitted modal pressures.
 $[T]_{N \times N}$ = The multimodal transmission matrix
 $[S]_{N \times N}$ = The scattering matrix
 $\{P^{\text{out}}\}_{2N}$ = The outgoing pressure wave vector
 $\{P^{\text{in}}\}_{2N}$ = The incoming pressure wave vector
 $[I]_{N \times N}$ = The identity matrix
 M = Number of source configurations
 $\Gamma_{\text{WD}}, \vec{n}_{\text{WD}}$ = The rigid walls and its normal vector
 $\Gamma_{\text{LD}}, \vec{n}_{\text{LD}}$ = The lined walls and its normal vector
 Ω = The acoustic domain inside the duct
 Z = Acoustic impedance
 $\Gamma_L, \vec{n}_L, \Gamma_R$ and $\vec{n}_R$ = Left and right boundaries and their normal vectors
 Π = The weak variational formulation
 q = The test function
 $\{p\}$ = The nodal acoustic pressure vector
 $[\Delta]$ = The azimuthal scattering matrix
 W_{dis} = The acoustic power dissipation
 W^{in} = The total acoustic power of incoming waves
 W^{out} = The total acoustic power of outgoing waves.
 F_{obj} = The cost function
 e = Thickness of the perforated facing sheet
 D = Hole diameter of the perforated sheet.
 σ = Open area of the perforated facing sheet
 ρ_0 = The air mass per volume unit

APPENDIX

The total acoustic power of incoming waves W^{in} and the total acoustic power of outgoing waves W^{out} are expressed as follows:

$$\begin{aligned}
 W^{\text{in}} &= \sum_{m=-P}^P \sum_{n=0}^Q \frac{N_{mn} \cdot Y_{mn}}{2} (|P_{mn}^{\text{I+}}(z_L)|^2 + |P_{mn}^{\text{II-}}(z_R)|^2), \\
 W^{\text{out}} &= \sum_{m=-P}^P \sum_{n=0}^Q \frac{N_{mn} \cdot Y_{mn}}{2} (|P_{mn}^{\text{I-}}(z_L)|^2 + |P_{mn}^{\text{II+}}(z_R)|^2). \quad (\text{A1})
 \end{aligned}$$

These total powers can be written in the following form:

$$\begin{aligned}
 W^{\text{in}} &= \{\Pi^{\text{in}}\}_{2N}^{T*} \cdot \{\Pi^{\text{in}}\}_{2N}, \\
 W^{\text{out}} &= \{\Pi^{\text{out}}\}_{2N}^{T*} \cdot \{\Pi^{\text{out}}\}_{2N}, \quad (\text{A2})
 \end{aligned}$$

where

$$\begin{aligned}
 \{\Pi^{\text{in}}\}_{2N}^T &= \langle \Pi_{00}^{\text{I+}}(z_L), \dots, \Pi_{PQ}^{\text{I+}}(z_L), \Pi_{00}^{\text{II-}}(z_R), \dots, \Pi_{PQ}^{\text{II-}}(z_R) \rangle, \\
 \{\Pi^{\text{out}}\}_{2N}^T &= \langle \Pi_{00}^{\text{I-}}(z_L), \dots, \Pi_{PQ}^{\text{I-}}(z_L), \Pi_{00}^{\text{II+}}(z_R), \dots, \Pi_{PQ}^{\text{II+}}(z_R) \rangle, \\
 \Pi_{mn}^{\text{I+}}(z_L) &= X_{mn} P_{mn}^{\text{I+}}(z_L), \quad \Pi_{mn}^{\text{II+}}(z_R) = X_{mn} P_{mn}^{\text{II+}}(z_R), \quad X_{mn} = \sqrt{N_{mn} k_{mn} / 2\rho_0 c_0 k}, \text{ and } \rho_0 \text{ the air mass per volume unit. In a} \\
 &\text{matrix form, we have the following relations:}
 \end{aligned}$$

$$\{\Pi^{\text{in}}\}_{2N} = [X]_{2N \times 2N} \cdot \{P^{\text{in}}\}_{2N}, \quad \{\Pi^{\text{out}}\}_{2N} = [X]_{2N \times 2N} \cdot \{P^{\text{out}}\}_{2N}, \quad (\text{A3})$$

where

$$[X]_{2N \times 2N} = \begin{bmatrix} [\text{diag}(X_{mn})]_{N \times N} & [0]_{N \times N} \\ [0]_{N \times N} & [\text{diag}(X_{mn})]_{N \times N} \end{bmatrix}. \quad (\text{A4})$$

The scattering matrix relates the incoming pressure waves to outgoing pressure waves:

$$\begin{aligned}
 \{\Pi^{\text{out}}\}_{2N} &= [S']_{2N \times 2N} \cdot \{\Pi^{\text{in}}\}_{2N} \\
 &= [X]_{2N \times 2N} [S]_{2N \times 2N} [X]_{2N \times 2N}^{-1} \cdot \{\Pi^{\text{in}}\}_{2N}. \quad (\text{A5})
 \end{aligned}$$

The acoustic power dissipation becomes

$$W_{\text{dis}} = \{\Pi^{\text{in}}\}_{2N}^{T*} [[I]_{2N \times 2N} - [H]_{2N \times 2N}]_{2N \times 2N} \cdot \{\Pi^{\text{in}}\}_{2N}, \quad (\text{A6})$$

where

$$[H]_{2N \times 2N} = [S']_{2N \times 2N}^{T*} \cdot [S']_{2N \times 2N}. \quad (\text{A7})$$

This matrix can be expressed as

$$[H]_{2N \times 2N} = [U]_{2N \times 2N} \cdot [\Lambda]_{2N \times 2N} \cdot [U]_{2N \times 2N}^{T*}, \quad (\text{A8})$$

where $[U]$ and $[\Lambda]$ are, respectively, the matrices of eigenvectors and eigenvalues of $[H]$. The expression (37) becomes

$$W_{\text{dis}} = \{d\}_{2N}^{T*} \cdot [\text{diag}(1 - \lambda_i)]_{2N \times 2N} \cdot \{d\}_{2N}, \quad (\text{A9})$$

where λ_i are the eigenvalues of the matrix $[H]$ and $\{d\}_{2N} = [U]_{2N \times 2N}^{T*} \cdot \{\Pi^{\text{in}}\}_{2N}$.

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